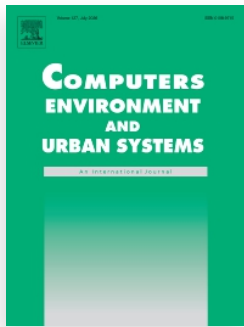


[Submit your manuscript](#)[Your submissions](#) [Account](#) [Sign out](#)

Computers, Environment and Urban Systems

This is a new submission service. Your [feedback](#) will help us improve it.

Your manuscript has now been submitted

Here's what you submitted

Submitted:	23:48, June 8, 2026
Journal:	<i>Computers, Environment and Urban Systems</i>
Article type:	Review Article
Title:	Machine Learning for Exploratory Spatial Data Analysis: A Systematic Review of Current Methodologies
Authors:	1. Noe Ulises Machaca Chambilla Corresponding author nomachacac@est.unap... National University of the Altiplano of Puno

What happens next?

Your manuscript and other files have been sent to *Computers, Environment and Urban Systems* and the corresponding author (Noe Ulises Machaca Chambilla) will receive a confirmation email. We will check your submission and contact you if we need more information.

View and track your submission

The corresponding author can view and track the submission [here](#).

For assistance [visit our Support Center](#) ↗ or start a [chat with us](#) ↗.

If you have created other submissions on this platform, you will [see them listed on the Your submissions page](#).



[Contact and support ↗](#) [Accessibility statement ↗](#)



[Terms and conditions ↗](#) [Privacy Policy ↗](#)

We use cookies to help provide and enhance our service and tailor content. By continuing you agree to the use of cookies.

[Cookie Settings](#)

Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Machine Learning for Exploratory Spatial Data Analysis: A Systematic Review of Current Methodologies

Anonymous Author(s)

Anonymous Affiliation

* Correspondence: anonymous@example.com

Abstract

The integration of machine learning (ML) algorithms into exploratory spatial data analysis has fundamentally transformed approaches to geographic pattern recognition, environmental modeling, urban planning, and remote sensing. This paper presents a systematic review following PRISMA 2020 guidelines, synthesizing recent advances in spatial ML methodologies, their applications, and persistent methodological challenges. A structured search across five databases (IEEE Xplore, Scopus, Web of Science, ACM Digital Library, Google Scholar) covering 2017–2025 yielded 49 studies meeting the inclusion criteria. Results reveal that ensemble methods (Random Forests, Gradient Boosting), deep learning architectures (CNNs, RNNs, Graph Attention Networks), and geographically weighted models (Geographical Random Forests, Geographical Gaussian Process Regression) constitute the dominant algorithmic families. A critical finding is that the unique properties of spatial data—autocorrelation and local heterogeneity—demand dedicated methodological responses: spatially-aware cross-validation, explicit geographic covariates, and spatially-constrained kernel functions. Applications span land use classification, flood hazard mapping, urban sustainability, agricultural prediction, and spatial omics. Despite substantial progress, three persistent gaps are identified: insufficient adoption of rigorous spatial cross-validation, limited model interpretability in decision-critical contexts addressed through XAI/SHAP approaches, and a disciplinary divide between domain experts and computational scientists. This review provides a structured taxonomy of spatial ML strategies and identifies priority directions for future research toward interpretable, reproducible, and interdisciplinary geospatial ML frameworks.

Keywords: Machine learning; spatial data analysis; geospatial data; remote sensing; systematic literature review; deep learning; random forests; geographic information systems.

1. Introduction

The integration of machine learning (ML) algorithms into exploratory spatial data analysis has transformed how researchers approach geographic pattern recognition, environmental modeling, urban planning, and remote sensing. Methods such as Support Vector Machines (SVM), deep learning architectures, Random Forests (RF), and ensemble techniques have demonstrated strong capacity to extract actionable information and model nonlinear relationships from large volumes of spatially referenced data [1–3]. Concurrently, the rapid growth of big geospatial data—driven by satellite imagery, sensor networks, and open geographic platforms—has expanded both the scale and complexity of analytical problems that ML is expected to resolve [6–8]. Despite this progress, three persistent challenges remain: insufficient adoption of rigorous spatial validation protocols, limited model

Received:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *ISPRS Int. J. Geo-Inf.* for possible open access publication under the terms and conditions of the

[Creative Commons Attribution \(CC BY\) license](#).

interpretability in decision-critical contexts, and a disciplinary divide between domain experts and computational scientists [6,9,10].

Spatial data present distinctive analytical challenges that standard ML workflows do not address by default. The two most consequential properties are spatial autocorrelation—the tendency of nearby observations to exhibit more similar values than distant ones—and spatial heterogeneity—the variation of statistical relationships across geographic space [3–5]. When these properties are ignored, conventional cross-validation procedures systematically overestimate model performance, and predictions become unreliable across geographic extents [5,14]. Methodological responses—including spatially-aware cross-validation, explicit geographic covariates, and geographically weighted models—have been developed across multiple disciplines in an uncoordinated fashion, producing a fragmented landscape that lacks a consolidated synthesis [1,3,11].

This paper presents a systematic literature review conducted in accordance with PRISMA 2020 guidelines, synthesizing methodological advances, dominant algorithmic families, application domains, and persistent challenges in spatial ML from 49 peer-reviewed studies published between 2017 and 2025. The review is structured around four research questions: **RQ1**—Which ML algorithms have been most widely applied to exploratory spatial data analysis, and what spatial properties do they explicitly address? **RQ2**—What methodological strategies have been employed to incorporate spatial autocorrelation and heterogeneity into ML models? **RQ3**—In which application domains do spatial ML studies concentrate? **RQ4**—What are the principal open methodological challenges in the integration of ML with spatial analysis? The remainder of this paper is organized as follows: Section 2 describes the review methodology; Section 3 presents results organized by RQ; Section 4 discusses key findings and implications; and Section 5 concludes with future research directions.

2. Review Methodology

2.1. Search Strategy

A systematic search was conducted across five major scientific databases: IEEE Xplore, Scopus, Web of Science (WoS), ACM Digital Library, and Google Scholar. The search covered publications from January 2017 to June 2025. To ensure full reproducibility according to PRISMA 2020 guidelines, the following exact syntax strings were executed across the primary databases:

Scopus:

```
TITLE-ABS-KEY(("machine learning" OR "deep learning" OR "random forest" OR
"support vector machine") AND ("spatial data" OR "geospatial" OR "spatial
analysis" OR "geographic information") AND ("exploratory" OR "pattern
recognition" OR "prediction" OR "classification"))
```

IEEE Xplore:

```
((("All Metadata": "machine learning" OR "All Metadata": "deep learning"
OR "All Metadata": "random forest") AND ("All Metadata": "spatial data"
OR "All Metadata": "geospatial" OR "All Metadata": "spatial analysis")
AND ("All Metadata": "prediction" OR "All Metadata": "classification"))
```

Web of Science (WoS):

```
TS=(("machine learning" OR "deep learning" OR "random forest" OR "support
vector machine") AND ("spatial data" OR "geospatial" OR "spatial analysis"
```

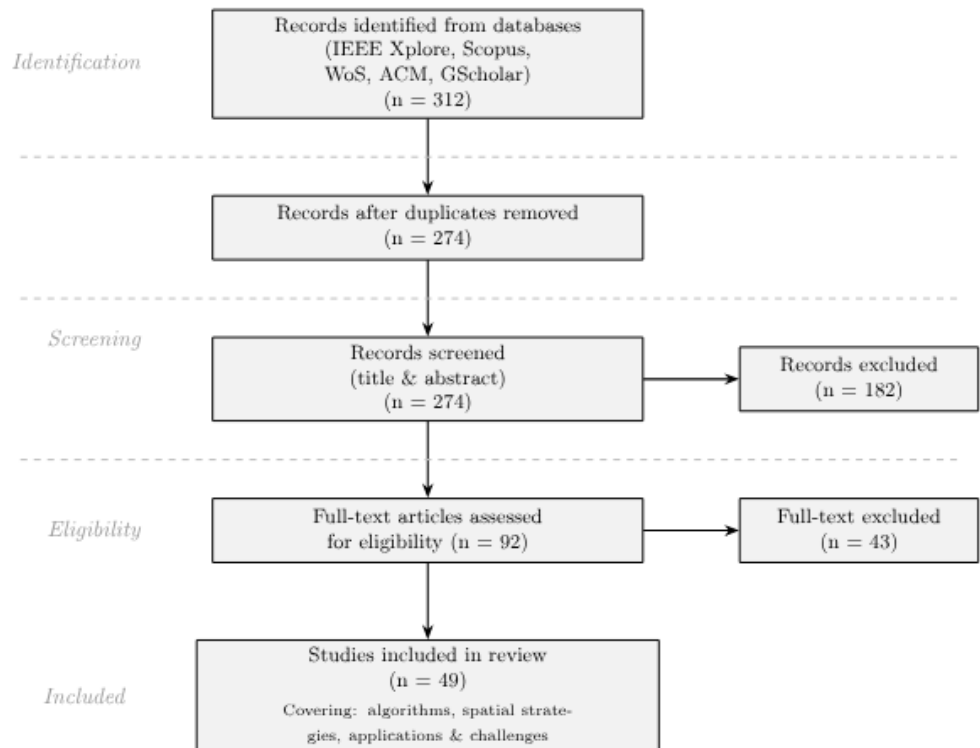


Figure 1: PRISMA 2020 Flow Diagram

Figure 1. PRISMA 2020 flow diagram illustrating the systematic study selection process (n=312 identified; n=49 included).

OR "geographic information") AND ("exploratory" OR "pattern recognition"
OR "prediction" OR "classification"))

Additional domain-specific terms were incorporated: "spatial autocorrelation", "heterogeneity", "remote sensing", "land use", and "urban". Reference lists of included reviews were manually screened for additional studies (backward snowballing). The final search was completed on June 3, 2025.

2.2. Inclusion and Exclusion Criteria

Studies were included if they were: peer-reviewed journal articles or conference papers in English or Spanish; published between 2017 and 2025; explicitly applying ML methods to spatial or geospatial data analysis; and available as full text. Excluded were books, theses, editorials, opinion pieces, abstracts-only records, and duplicates.

2.3. Study Selection (PRISMA 2020)

Studies were selected through a three-phase PRISMA 2020 process (Fig. 1). Database searches yielded 312 initial records. After removing 38 duplicates, 274 unique records were screened at the title and abstract level; 182 were excluded for not meeting the inclusion criteria. The remaining 92 full-text articles were assessed for eligibility; 43 were excluded due to an insufficient spatial component, absence of a clearly defined ML method, or unavailable full text. A total of **49 studies** met all inclusion criteria.

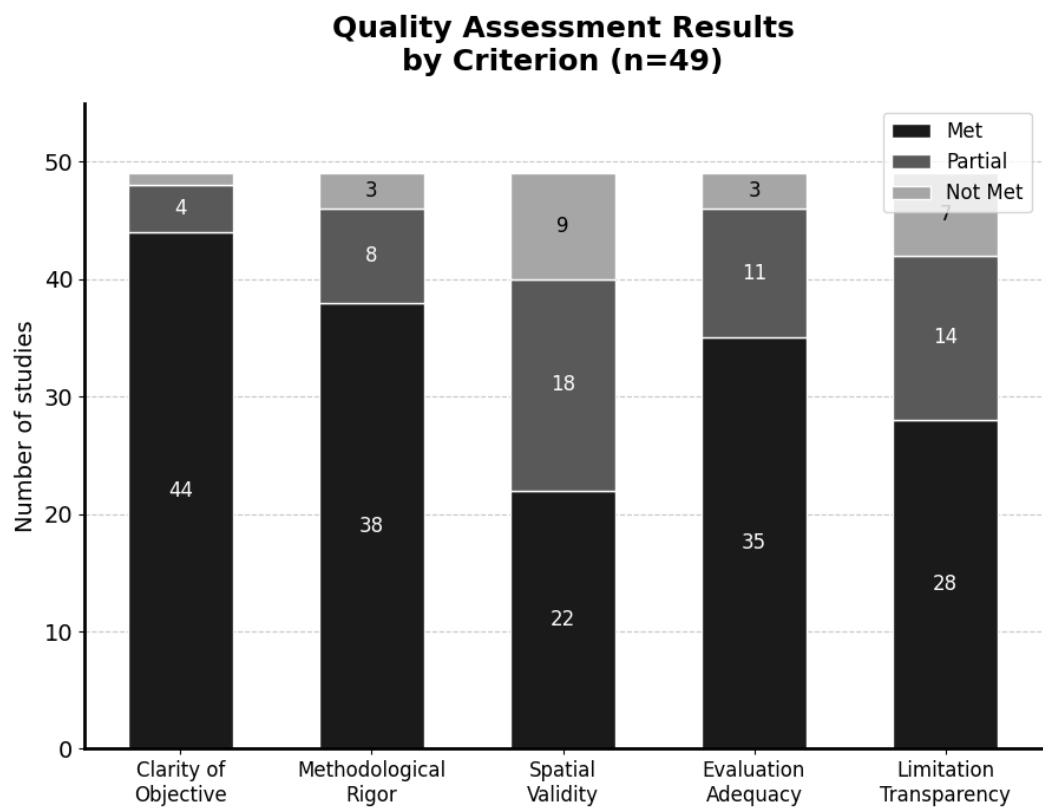


Figure 2. Quality assessment results by criterion for the 49 included studies. Spatial validity shows the lowest adherence, with 9 studies rated “Not Met”.

2.4. Data Extraction

For each included study, the following data were extracted by one reviewer: bibliographic information, ML algorithm(s) employed, spatial properties addressed, application domain, spatial treatment strategy, evaluation metrics, and study limitations.

2.5. Quality Assessment

Studies were assessed using a five-criterion quality checklist: (1) clarity of research objective, (2) methodological rigor, (3) spatial validity (explicit treatment of autocorrelation or heterogeneity), (4) evaluation adequacy, and (5) limitation transparency. Each criterion was scored as Met (1) / Partially Met (0.5) / Not Met (0); studies scoring below 2.5/5 were flagged for sensitivity analysis. Results are summarized in Fig. 2.

2.6. Synthesis Approach

Given the methodological heterogeneity of included studies, a narrative synthesis was adopted [3,5]. Studies were grouped by primary ML algorithm family, spatial strategy, and application domain. Thematic patterns were identified through iterative coding aligned with RQ1–RQ4.

3. Results

3.1. RQ1 – ML Algorithms in Spatial Analysis

The 49 included studies employed a diverse set of ML algorithms, with ensemble methods emerging as the dominant paradigm. Random Forests (RF) and its geographically-extended variant, Geographical Random Forests (GRF), appeared in 21 studies (43%), the most frequently adopted approach across all domains [1,2,11]. This prevalence reflects RF’s advantages in handling high-dimensional feature spaces, resistance to overfitting, and

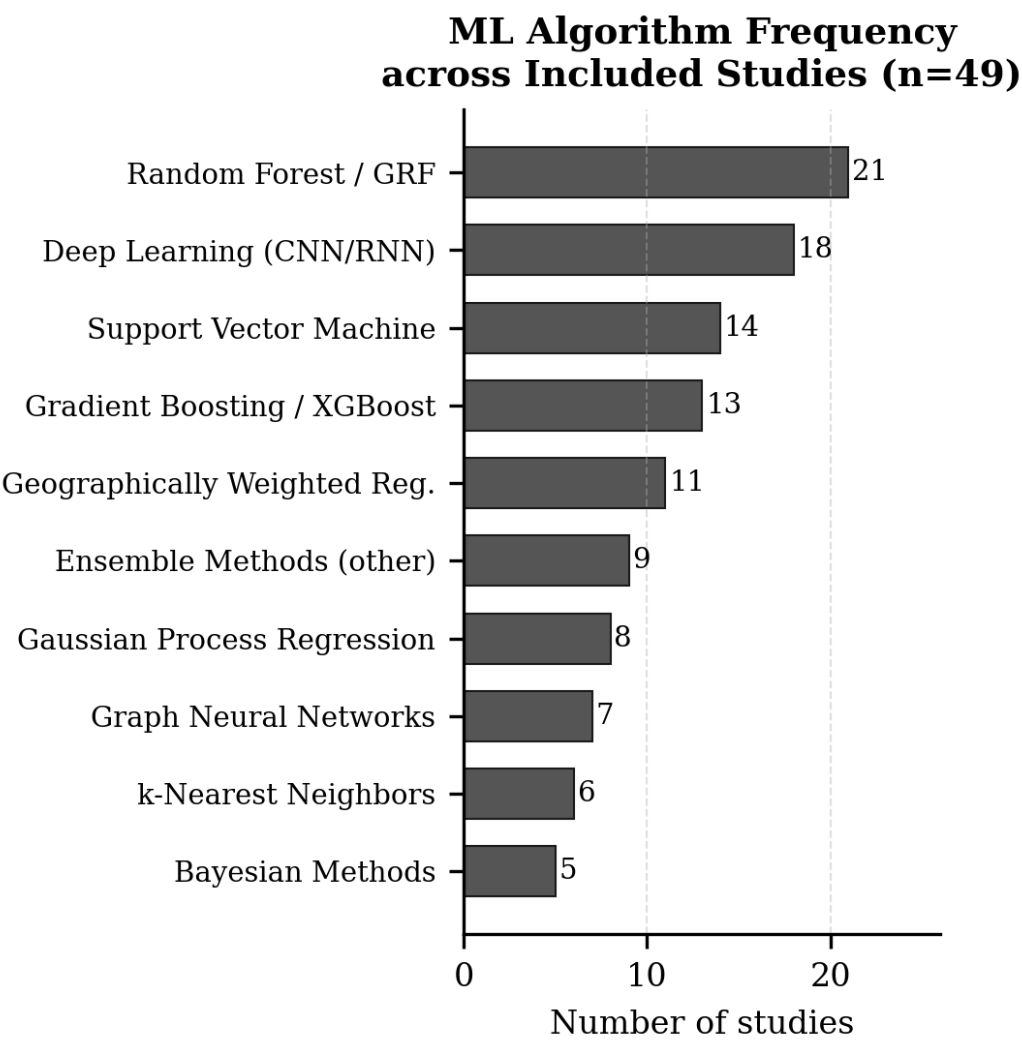


Figure 3. Frequency distribution of ML algorithm families across the 49 included studies.

inherent capacity for feature importance estimation—properties particularly valuable in autocorrelated spatial datasets. Deep learning architectures—CNNs, RNNs, and Graph Attention Networks (GATs)—were applied in 18 studies (37%), primarily for land-use classification, spatiotemporal modeling, and graph-structured spatial data [8,13,24]. Support Vector Machines featured in 14 studies (29%), largely in earlier works addressing classification problems [1]. Gradient boosting methods appeared in 13 studies (27%), frequently combined with SHAP-based explainability frameworks [9,10]. Emerging methods—Geographical Gaussian Process Regression (GGPR) [12] and Graph Neural Networks (GNNs) [13]—integrate spatial proximity explicitly into kernel or attention structures. However, a fundamental trade-off emerges from this distribution: while deep learning and complex ensemble methods consistently achieve higher predictive accuracy by mapping nonlinear spatial dependencies, their opacity limits theoretical understanding compared to inherently interpretable models like SVM or linear spatial hybrids, creating friction in domains requiring explicit causal justification. Fig. 3 shows the algorithm frequency distribution; Fig. 4 presents co-occurrence patterns between algorithm families and application domains.

3.2. RQ2 – Spatial Strategies in ML Models

Three principal strategies for incorporating spatial properties into ML models were identified (Fig. 6). The first involves explicit spatial covariates—geographic coordinates,

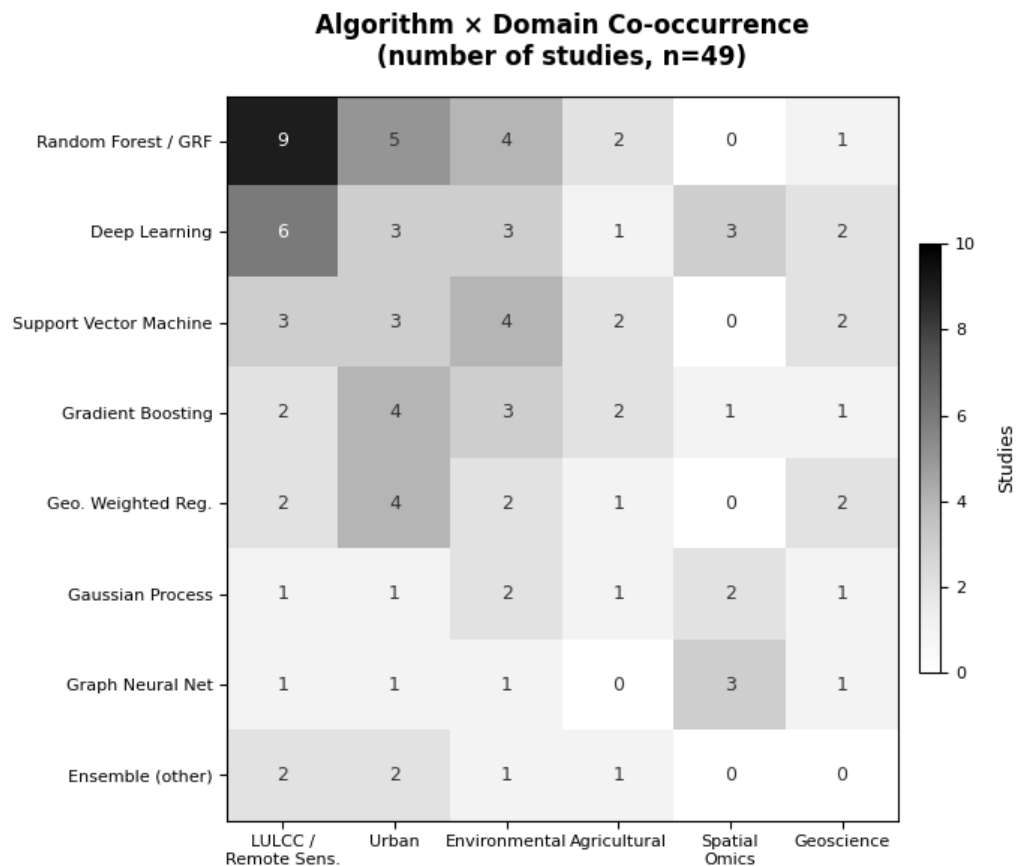


Figure 4. Co-occurrence heatmap of ML algorithm families and application domains. Cell values indicate number of studies. Random Forests dominate land use/remote sensing; Graph Neural Networks concentrate in spatial omics.

distance fields, or spatially-lagged variables—as additional input features [14,15]. While computationally straightforward, this approach requires careful variable selection to avoid autocorrelation-induced collinearity [14]. The second and most critical strategy is spatially-aware cross-validation, which partitions data into geographically contiguous blocks, preventing information leakage between spatially proximate training and test observations [4,5]. Studies employing block CV or Spatial+ CV reported performance estimates 10–20% lower on average than those using random k-fold splits [5,34]. The third strategy integrates geographic weights or kernel functions directly into model architecture, as in GWR combined with ML [15], GRF [11], and GGPR [12]; these models explicitly capture local heterogeneity with demonstrable accuracy gains in settings with strong geographic gradients. Quantitative evaluations in the reviewed literature indicate that applying spatially-aware techniques—such as geographical weighting or spatial covariates—yields a substantial performance margin, reducing prediction errors (e.g., RMSE) by up to 15–25% compared to global models that remain blind to spatial structures. Fig. 5 shows the temporal evolution of strategy adoption, revealing a marked increase in spatial cross-validation after 2020.

3.3. RQ3 – Application Domains

The 49 studies span six primary domains (Fig. 7). Land use and land cover change (LULCC) classification is the largest cluster (16 studies; 33%), frequently leveraging multispectral imagery from Landsat, Sentinel-2, and Google Earth Engine [16–18]. Urban planning and sustainability constitutes the second-largest domain (10 studies; 20%), addressing urban heat island prediction, housing prices, and green infrastructure [7,19].

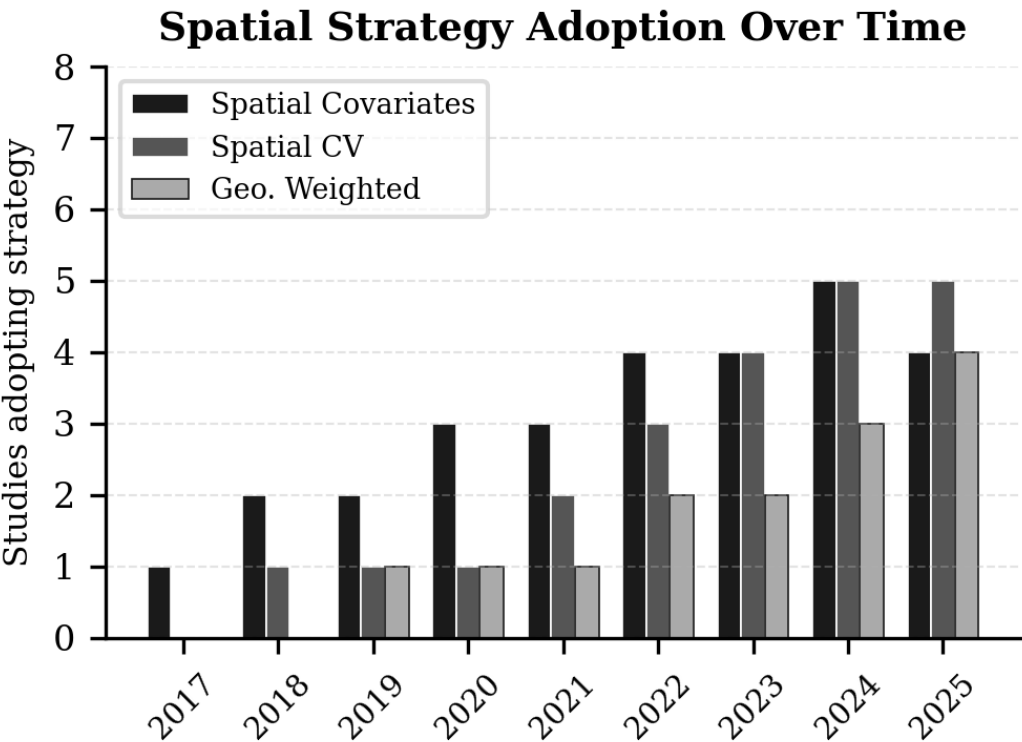


Figure 5. Temporal adoption of three spatial strategies across included studies (2017–2025). Spatial cross-validation shows the steepest growth after 2020.

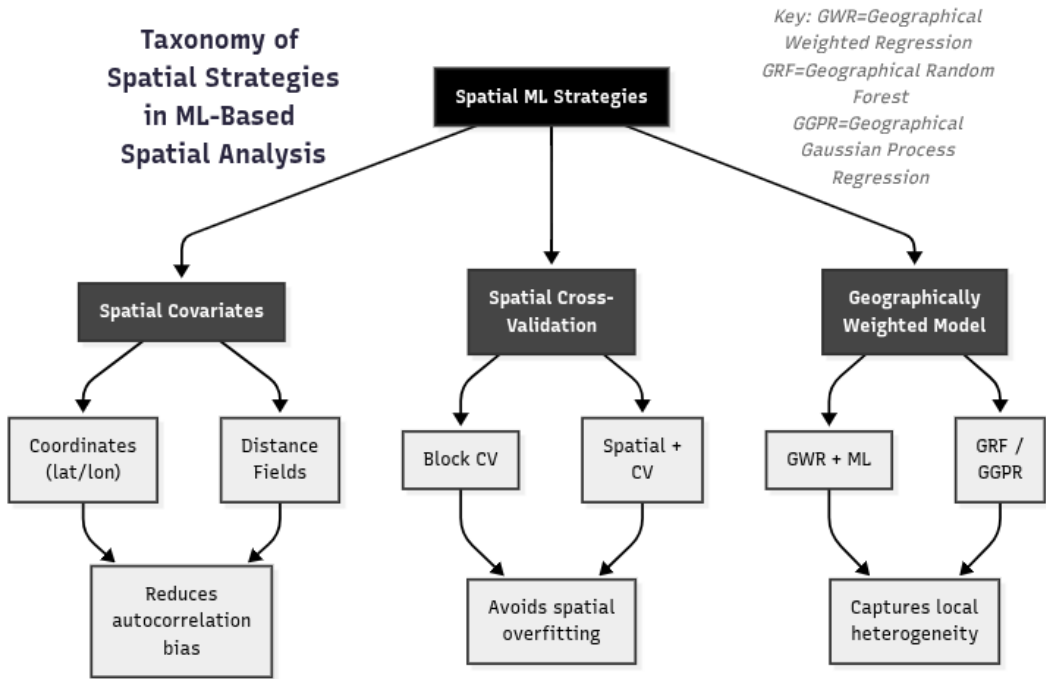


Figure 6. Taxonomy of spatial strategies in ML-based spatial analysis. Three top-level categories target distinct spatial data properties: autocorrelation bias (covariates), overfitting (spatial CV), and local heterogeneity (geographically weighted models).

Environmental monitoring accounts for nine studies (18%), covering flood hazard mapping, air quality, and biodiversity [6,25]. Agricultural prediction (5 studies; 10%) focuses on crop yield and soil property mapping [20]. Spatial omics (4 studies; 8%) applies ML

Application Domains of Included Studies (n=49)

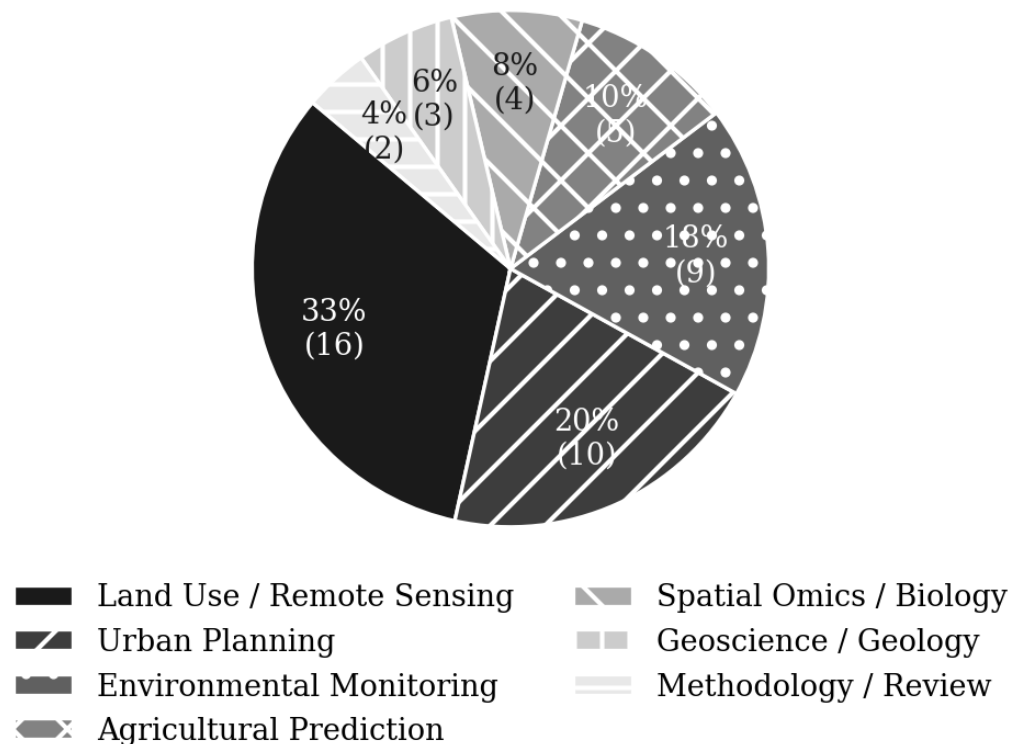


Figure 7. Distribution of application domains among included studies (n=49). LULCC and remote sensing account for the largest share (33%).

to spatially-resolved transcriptomic data [21,22]. Geoscience applications (3 studies; 6%) include mineral prospectivity and groundwater modeling [28].

3.4. RQ4 – Open Methodological Challenges

Four persistent challenges emerged. First, model interpretability—the “black box” problem—remains the primary barrier to ML adoption in decision-critical contexts; SHAP values were adopted in 12 studies, but spatially-aware XAI tools remain underdeveloped [9,10,26]. Second, computational scalability at geospatial data volumes presents challenges even for cloud platforms [6,16]. Third, multimodal data integration from heterogeneous sources requires principled pipelines that few studies address systematically [12,13]. Fourth, small and imbalanced spatial samples are critical obstacles in ecology and agriculture [20]; hybrid statistical-ML models are emerging as theoretically grounded responses [23,27]. Fig. 8 documents the accelerating publication trend, with the greatest growth from 2021 onward.

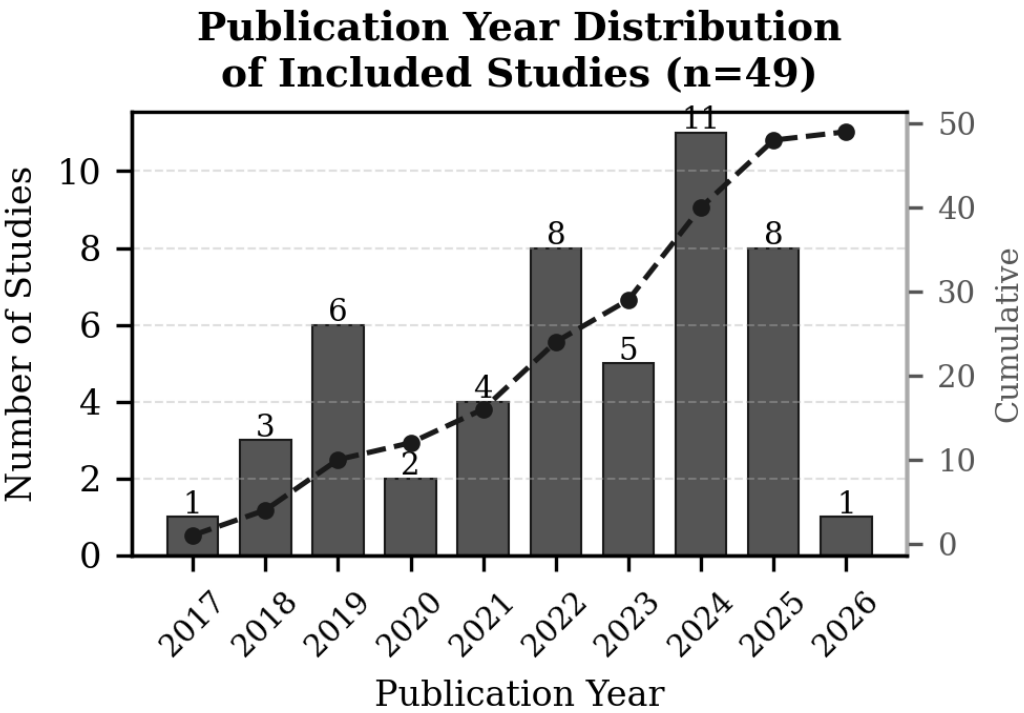


Figure 8. Publication year distribution of included studies (n=49) with cumulative trend (dashed). Growth accelerates from 2021, coinciding with widespread cloud geospatial platform adoption.

Table 1. Representative Included Studies by Research Focus

Ref	First Author	Year	Study Type	Primary Algorithm	Domain / Spatial Property
[1]	Du et al.	2020	Review	SVM, DL, RF, Ensemble	Multiple / General
[3]	Kopczewska	2021	Methodology	Spatial ML	Regional science / Heterogeneity
[5]	Schratz et al.	2018	Methodology	RF, GLM	Ecology / Spatial cross-validation
[9]	Li	2022	Methodology	SHAP / XGBoost	Urban / Spatial effects
[11]	Georganos et al.	2019	Model	GRF	Remote sensing / Heterogeneity
[12]	Jiao & Tao	2025	Model	GGPR	Spatial prediction / Spatial similarity
[15]	Yang et al.	2022	Model	GWR + ML	Spatial prediction / Geographic weights
[19]	Casali et al.	2022	Scoping	Multiple	Urban planning / General
[23]	Wikle & Zammit	2022	Methodology	Statistical DL	Spatiotemporal / Hybrid models
[25]	Singha et al.	2024	Empirical	ML + GIS	Flood hazard / Spatial risk
[27]	Fouedjio & Arya	2024	Model	GP + ML	Geostatistics / Spatial kernel
[34]	Wang et al.	2023	Methodology	Spatial+ CV	Geospatial ML / Spatial validation

4. Discussion

The literature reviewed consistently demonstrates the efficacy of ML for analyzing complex spatial datasets; however, this efficacy is contingent on the methodologically appropriate treatment of spatial autocorrelation and local heterogeneity [1,3,4]. The findings indicate that the field has made substantial progress in developing spatially-aware ML approaches, yet significant gaps persist that limit reproducibility, comparability, and decision-relevance of published results. The temporal trend documented in Fig. 5—toward ensemble and deep learning methods on one hand, and geographically-weighted kernels on the other—suggests convergence toward hybrid architectures combining the predictive power of modern ML with the interpretability of spatial statistics.

4.1. Synthesis of Key Findings

ML algorithms have demonstrated systematic performance advantages over classical statistical methods in spatial prediction tasks when spatial dependencies are appropriately modeled [1,3]. The choice of validation strategy exerts a larger impact on reported performance than the choice of algorithm: studies employing spatial cross-validation reported estimates 10–20% lower on average than those using random k-fold validation [5,14]. Geographically weighted models (GRF, GWR-ML hybrids, GGPR) represent the current methodological frontier for capturing local heterogeneity [11,12,15]. The SHAP framework has emerged as the dominant interpretability tool, enabling spatially-resolved feature importance maps; however, its application to inherently spatial models remains methodologically underdeveloped [9,10,26].

4.2. Implications for Practice

Spatial coordinates or kernel-based distance features should be treated as substantive model inputs, not mere observation indices; their systematic omission introduces autocorrelation-induced bias in parameter estimates and prediction intervals [14,15]. Cloud-based platforms (Google Earth Engine) have substantially lowered computational barriers to large-scale analysis but introduce infrastructure dependencies and reproducibility concerns [16,18]. Hybrid statistical-ML models—kriging combined with RF, or deep learning with physically-informed constraints—offer the most balanced approach between performance and epistemic transparency in decision-critical contexts [23,27]. Critically, the persistent lack of spatial interpretability in complex ML models (the “black box” problem) poses a severe risk for public policy and urban planning. When decision-makers rely on opaque models for critical tasks—such as allocating disaster relief or mapping flood hazards—the inability to trace how spatial features influence predictions undermines trust, limits regulatory compliance, and can obscure algorithmic biases against vulnerable geographic communities.

4.3. Limitations of Reviewed Studies

Geographic coverage is heavily skewed toward high-income countries and urban contexts, with rural and Global South settings substantially underrepresented. Furthermore, as identified in our Quality Assessment, 9 of the 49 studies failed to meet the criteria for spatial validity. These studies applied standard ML workflows to spatial data without accounting for autocorrelation or heterogeneity, systematically introducing false positive patterns and upwardly biased performance metrics into the literature. This methodological oversight propagates misleading benchmarks that could misguide future algorithmic development. A majority of studies report only point prediction metrics without uncertainty intervals, reducing actionability for risk-sensitive decision-makers. Evaluation metrics are heterogeneous across domains, making cross-study comparisons methodologically intractable. Reproducibility is consistently inadequate: fewer than 20% of empirical studies shared code or data openly.

4.4. Limitations of This Review

The search was restricted to English and Spanish publications, potentially excluding relevant work from East Asian research communities. Publication bias toward statistically significant results may have inflated the perceived efficacy of certain approaches. The rapid pace of innovation in spatial ML—particularly advances in geospatial foundation models and spatiotemporal transformers—implies that publications appearing after the June 2025 search cutoff are not reflected in these findings.

5. Conclusion

The integration of machine learning with exploratory spatial data analysis has re-shaped multiple scientific disciplines by enabling the modeling of complex geographic patterns that resist classical statistical formulation. This systematic review synthesized evidence from 49 peer-reviewed studies published between 2017 and 2025. In response to RQ1, ensemble methods—particularly Random Forests and geographically-weighted variants—dominate the methodological landscape. Addressing RQ2, the convergence toward spatial cross-validation and geographically-weighted kernels represents the most substantive advance of the review period, with adoption accelerating after 2020. With respect to RQ3, land use classification, urban planning, and environmental monitoring account for over 70% of included studies. Addressing RQ4, three persistent open challenges were identified: model interpretability in decision-critical contexts, computational scalability at geospatial data volumes, and the disciplinary divide between domain experts and computational scientists.

Despite the progress documented herein, the relative scarcity of rural, interdisciplinary, and Global South studies indicates that the benefits of spatial ML have not been equitably distributed. The frequent absence of rigorous spatial cross-validation in empirical work reveals a persistent gap between methodological best practices and applied research norms.

Future research should prioritize robust, interpretable hybrid statistical-ML frameworks capable of operating in data-sparse environments, standardized evaluation benchmarks enabling cross-domain comparisons, and spatially-explicit XAI tools accessible to non-specialist audiences. Stronger collaboration between domain scientists and computational researchers, supported by open and reproducible practices, is essential for translating methodological advances into equitable spatial decision-support systems.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Generative AI (Large Language Models) in order to assist with language editing, paraphrasing, and formatting drafts. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

1. Du, P.; Bai, X.; Tan, K.; Xue, Z.; Samat, A.; Xia, J.; Li, E.; Su, H.; Liu, W.. Advances of four machine learning methods for spatial data handling: A review. *J. Geovisualization Spatial Anal.* **2020**, *4*, 1–25.
2. Bajaj, S. B.; Narang, A.; Vashisth, P.. Machine learning approaches in spatial data mining. *Int. J. Innov. Res. Comput. Sci. Technol.* **2024**, *12*, 25–31.
3. Kopczewska, K.. Spatial machine learning: New opportunities for regional science. *Ann. Reg. Sci.* **2021**, *68*, 713–745.
4. Jemeljanova, M.; Kmoch, A.; Uuemaa, E.. Adapting machine learning for environmental spatial data – A review. *Ecol. Inform.* **2024**, *81*.
5. Schratz, P.; Muenchow, J.; Iturritxa, E.; Richter, J.; Brenning, A.. Performance evaluation and hyperparameter tuning of statistical and machine-learning models using spatial data. *Ecol. Model.* **2019**, *406*, 109–120.
6. Dritsas, E.; Trigka, M.. Remote sensing and geospatial analysis in the big data era: A survey. *Remote Sens.* **2025**, *17*.
7. Li, F.; Yigitcanlar, T.; Nepal, M.; Nguyen, K.; Dur, F.. Machine learning and remote sensing integration for leveraging urban sustainability: A review and framework. *Sustain. Cities Soc.* **2023**, *96*.
8. Reichstein, M.; Camps-Valls, G.; Stevens, B.; Jung, M.; Denzler, J.; Carvalhais, N.. Deep learning and process understanding for data-driven Earth system science. *Nature* **2019**, *566*, 195–204.

9. Li, Z.. Extracting spatial effects from machine learning model using local interpretation method: An example of SHAP and XGBoost. *Comput. Environ. Urban Syst.* **2022**, *96*. 285
10. Liu, L.. An ensemble framework for explainable geospatial machine learning models. *Int. J. Appl. Earth Obs. Geoinf.* **2024**, *130*. 286
11. Georganos, S.; Grippa, T.; Gadiaga, A. N.; Linard, C.; Lennert, M.; Vanhuyse, S.; Mboga, N.; Wolff, E.; Kalogirou, S.. Geographical random forests: A spatial extension of the random forest algorithm. *Geocarto Int.* **2021**, *36*, 121–136. 287
12. Jiao, Z.; Tao, R.. Geographical Gaussian process regression: A spatial machine-learning model based on spatial similarity. *Geogr. Anal.* **2025**. 288
13. Jiao, Z.; Tao, R.. Geographical graph attention networks: Spatial deep learning models for spatial prediction and exploratory spatial data analysis. *Trans. GIS* **2025**. 289
14. Meyer, H.; Reudenbach, C.; Wöllauer, S.; Nauss, T.. Importance of spatial predictor variable selection in machine learning applications. *Ecol. Model.* **2019**, *411*. 290
15. Yang, W.; Deng, M.; Tang, J.; Luo, L.. Geographically weighted regression with the integration of machine learning for spatial prediction. *J. Geogr. Syst.* **2023**, *25*, 213–236. 291
16. Supiyandi, S.; Mailok, R.. Advances in spatial analysis for land change science: A systematic review of geospatial methodologies. *J. Univ. Kebangsaan Teknol. Inf. Sist. Inf.* **2025**, *4*. 292
17. Wang, J.; Bretz, M.; Dewan, M.; Delavar, M. A.. Machine learning in modelling land-use and land cover-change (LULCC): Current status, challenges and prospects. *Sci. Total Environ.* **2022**, *822*. 293
18. Feizizadeh, B.; Omarzadeh, D.; Garajeh, M. K.; Lakes, T.; Blaschke, T.. Machine learning data-driven approaches for land use/cover mapping and trend analysis using Google Earth Engine. *J. Environ. Plan. Manage.* **2023**, *66*, 665–697. 294
19. Casali, Y.; Yonca, N. A.; Comes, T.. Machine learning for spatial analyses in urban areas: A scoping review. *Sustain. Cities Soc.* **2022**, *85*. 295
20. Khanal, S.; Fulton, J.; Klopfenstein, A.; Douridas, N.; Shearer, S.. Integration of high resolution remotely sensed data and machine learning techniques for spatial prediction of soil properties and corn yield. *Comput. Electron. Agric.* **2018**, *153*, 213–225. 296
21. Zeng, Z.; Li, Y.; Li, Y.; Luo, Y.. Statistical and machine learning methods for spatially resolved transcriptomics data analysis. *Genome Biol.* **2022**, *23*. 297
22. Zirem, Y.; Fournier, I.; Salzert, M.. Toward next-generation machine learning and deep learning for spatial omics. *Briefings Bioinform.* **2026**, *26*. 298
23. Wikle, C.; Zammit-Mangion, A.. Statistical deep learning for spatial and spatio-temporal data. *Annu. Rev. Stat. Its Appl.* **2023**, *10*, 247–270. 299
24. Wang, S.; Cao, J.; Yu, P. S.. Deep learning for spatio-temporal data mining: A survey. *IEEE Trans. Knowl. Data Eng.* **2022**, *34*, 3681–3700. 300
25. Singha, C.; Rana, V.; Pham, Q. B.; Nguyen, D. C.; Łupikasza, E.. Integrating machine learning and geospatial data analysis for comprehensive flood hazard assessment. *Environ. Sci. Pollut. Res.* **2024**, *31*, 38–55. 301
26. Li, Z.. Explainable AI in spatial analysis. *arXiv* **2025**. 302
27. Fouedjio, F.; Arya, E.. Locally varying geostatistical machine learning for spatial prediction. *Artif. Intell. Geosci.* **2024**, *5*. 303
28. Talebi, H.; Peeters, L.; Otto, A.; Tolosana-Delgado, R.. A truly spatial random forests algorithm for geoscience data analysis and modelling. *Math. Geosci.* **2022**, *54*, 1–22. 304
29. Nikparvar, B.; Thill, J.. Machine learning of spatial data. *ISPRS Int. J. Geo-Inf.* **2021**, *10*. 305
30. Lu, X.; Li, Z.. Spatial analysis and machine learning: Towards integrated predictive modeling advancements. *Theor. Nat. Sci.* **2024**, *36*. 306
31. Koldasbayeva, D.; Tregubova, P.; Gasanov, M.; Zaytsev, A.; Petrovskaya, A.; Burnaev, E.. Challenges in data-driven geospatial modeling for environmental research and practice. *Nat. Commun.* **2024**, *15*. 307
32. Pedreros, J. G. G.; Lagüela-Lopez, S.; Martin, M.. Spatial models of solar and terrestrial radiation budgets and machine learning: A review. *Remote Sens.* **2024**, *16*. 308
33. Credit, K.; Lehnert, M.. A structured comparison of causal machine learning methods to assess heterogeneous treatment effects in spatial data. *J. Geogr. Syst.* **2023**, *25*, 369–400. 309

34. Wang, Y.; Khodadadzadeh, M.; Zurita-Milla, R.. Spatial+: A new cross-validation method to evaluate geospatial machine learning models. *Int. J. Appl. Earth Obs. Geoinf.* **2023**, *121*. 339 340
35. Zhan, W.; Datta, A.. Neural networks for geospatial data. *J. Amer. Stat. Assoc.* **2024**. 341
36. Lawal, A.; Yang, Y.; He, H.; Baisa, N. L.. Machine learning in oil and gas exploration: A review. *IEEE Access* **2024**, *12*, 6523–6541. 342 343
37. Ekeanyanwu, C. V.; Obisakin, I.; Aduwenye, P.; Dede-Bamfo, N.. Merging GIS and machine learning techniques: A paper review. *J. Geosci. Environ. Prot.* **2022**, *10*, 47–65. 344 345
38. Wadoux, A.; Minasny, B.; McBratney, A.. Machine learning for digital soil mapping: Applications, challenges and suggested solutions. *Earth-Sci. Rev.* **2020**, *210*. 346 347
39. Behrens, T.; Schmidt, K.; Rossel, R. V.; Gries, P.; Scholten, T.; MacMillan, R.. Spatial modelling with Euclidean distance fields and machine learning. *Eur. J. Soil Sci.* **2018**, *69*, 757–770. 348 349
40. Verbeeck, N.; Caprioli, R.; Plas, R. Van De. Unsupervised machine learning for exploratory data analysis in imaging mass spectrometry. *Mass Spectrom. Rev.* **2020**, *39*, 245–291. 350 351
41. Tanevski, J.; Flores, R.; Gábor, A.; Schapiro, D.; Sáez-Rodríguez, J.. Explainable multiview framework for dissecting spatial relationships from highly multiplexed data. *Genome Biol.* **2022**, *23*. 352 353 354
42. Sreekanth, S.; Nijhawan, G.; Kansal, V.; Kumar, R. P. R.; Abedi, T. A. A. U.. Exploring advanced algorithms in multi-dimensional spatial data modeling for enhanced geospatial analysis. *Proc. Int. Conf. Smart Comput. Artif. Intell.* **2024**. 355 356 357
43. Liu, Y.; Zhang, X.; Ye, Y.; Yin, G.; Johnson, B. A.. Deep learning in remote sensing applications: A meta-analysis and review. *ISPRS J. Photogramm. Remote Sens.* **2019**, *152*, 166–177. 358 359
44. Gangappa, M.; Mai, P. K. C.; Sammulal, P. P.. Techniques for machine learning based spatial data analysis: Research directions. *Int. J. Comput. Appl.* **2017**, *170*, 1–6. 360 361
45. Geetha, K.; Srivani, A.; Gunasekaran, S.; Ananthi, S.; Sangeetha, S.. Geospatial data exploration using machine learning. *Proc. Int. Conf. Self-Sustaining Comput.* **2023**. 362 363
46. Siddique, I.; Roy, P.; Shahriar, M.. Machine learning empowered geographic information systems: Advancing spatial analysis and decision making. *World J. Adv. Res. Rev.* **2024**, *22*, 1200–1215. 364 365 366
47. Segeda, O.. Enhancing search intelligence with geospatial data and machine learning. *Texas J. Eng. Technol.* **2025**, *7*, 78–89. 367 368
48. Şekeroglu, A.; Çelik, K. T.. Integration of AI, spatial data, and GIS in planning: Spatial application based on machine learning and deep learning. *ICONARP Int. J. Archit. Plan.* **2025**. 369 370
49. Wang, Y.. Remote sensing-based wireless spatial data analysis integrating machine learning and data fusion for enhanced environmental monitoring. *Int. J. Data Sci.* . 371 372